

Derek Martin

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Personal Summary

Mechanical Engineering student (aerospace focus, physics minor) with strength in space systems thinking and hands-on mechanical design. Built Python automation tools and currently developing Python control software (and supporting CAD/integration) for an autonomous piano-player capstone (target delivery June 2026). Designed and analyzed a high-safety-factor aluminum winch stand to lift a 150 lb motor ~50 ft. Seeking remote part-time engineering work through June 2027 and launch/space roles starting 2027.

Education

B.S. in Mechanical Engineering (Focus: Aerospace Engineering, Minor: Physics)

Oregon State University | Corvallis, Oregon

B.S. Expected Graduation: June 2026

Master's in Engineering Expected Graduation: June 2027

GPA: 3.76 & Honor Roll Every Term

Relevant Coursework: Space Systems Engineering, Systems Dynamics & Control, Dynamics, Heat Transfer, Mechanics of Materials, Instrumentation

Professional Experience

Engineering Intern | Evensol Inc. - EDG-L (Landfill Gas Collection) | June 2024 - September 2024

- Conducted equipment inspections, troubleshooting, and preventative maintenance on critical systems.
- Utilized Autodesk Fusion 360 to design and simulate a hand crank project for motor maintenance, gaining skills in 3D modeling and design evaluation.
- Supported feasibility studies and project pricing; contributed to cost estimates and documentation.
- Developed and maintained data-tracking spreadsheets; used Excel for analysis and reporting to support operational decision-making.
- Gained hands-on experience with SCADA systems, transmitter calibration, and gas sampling techniques.
- Collaborated across disciplines on site projects (installations, troubleshooting, and infrastructure improvements) in a safety-focused environment.

Wildland Firefighter | USFS Sisters Ranger District | August 2022 - September 2023

- Implemented fire suppression techniques, including direct and indirect fire line construction, demonstrating strong problem-solving under pressure.
- Operated various firefighting equipment and maintained chainsaw certification for hazardous tree removal and vegetation clearing.
- Collaborated effectively within a high-stakes team environment, utilizing navigation and GPS tools for efficient wildfire suppression.

Technical Skills

Software: Python (automation/control), Microsoft Excel, 3D Modeling (Autodesk Fusion 360), SCADA

Safety Protocols: PPE, working at heights, emergency response

Project Management: Evaluation, scheduling, resource management, data analysis and documentation

Systems: Space systems fundamentals, instrumentation exposure, reliability & safety mindset

Certifications

Emergency Medical Technician (EMT) | Certified through Impact EMS | June 9, 2024

Achievements

- Awarded "Hardest Worker" in Calculus (2021)
- Team Captain for Nordic Skiing
- Medallion awarded by Bend Police Department for integrity and discipline (2021)

Projects

- Autonomous Piano Player (Capstone, 2025–2026): Developing Python control software for an electromechanical system; supporting CAD/integration as needed.
- Winch Lift Stand: Designed and analyzed an aluminum stand supporting a winch to raise a 150 lb motor ~50 ft with a high factor of safety.
- Food Stirrer Prototype: Performed stress analysis and supported iterative design for a 3D-printed liquid-food stirrer with a 4-person team.
- Python Automation: Built a "Jarvis" assistant integrating calendar + smart-home control + weather/maps services; developed a versioned chess bot (~1000 Elo).